Database Architecture

The database layer uses PostgreSQL with Prisma ORM for type-safe access and migration management.

Technology Stack

Component	Choice	Purpose
Database	PostgreSQL 14+	Primary data store
ORM	Prisma 5	Type-safe queries and migrations
Migrations	Prisma Migrate	Schema versioning
Seed	Prisma Seed	Initial data population

Entity Relationship Diagram

Key Design Patterns

1. **Cascading Deletes:** Attachments, comments, history, and CSAT responses are deleted when their parent ticket is deleted.
2. **Soft References:** `TicketHistory.changedBy` and `KnowledgeArticle.createdBy` store user identifiers as strings, not foreign keys, preserving history even if users are deleted.
3. **JSON Fields:** `SavedView.filters`, `CustomFormField.options`, and `CustomFormField.conditions` use PostgreSQL JSON for flexible schema.
4. **Indexing:** Key fields like `email`, `username`, `ticketNumber`, `status`, `priority`, and `createdAt` are indexed for query performance.
5. **UUIDs:** All primary keys use UUIDs generated by the database for security and scalability.

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